Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose appropriate wave models and explain what a signal establishes.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two targets each have range 500 m but different radial velocities. Explain how they can have equal echo delays yet different Doppler shifts.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v = f\lambda$; $T = 1/f$; $d_{range} = v\,\Delta t/2$.
d_{range} is one-way range from round-trip time. Match speed and model to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

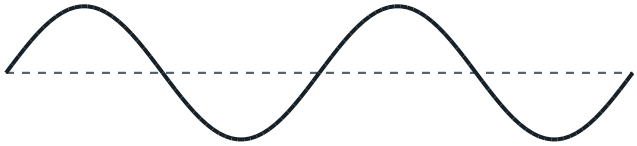
Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch a wave and label amplitude and wavelength 0. Explain which tells about oscillation size and which about spatial repetition.

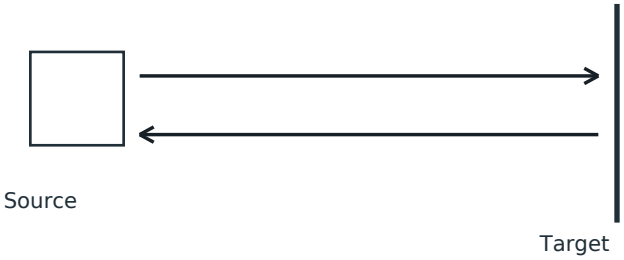


Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A radar receives an echo 0.012 s after transmission. Use $c = 3 \times 10^8$ m/s to find range. The radar frequency is 3×10^9 Hz; find wavelength in vacuum. Label the timing measurement and distinguish propagation speed from target motion.



Marzano check - circle: ① ② ③

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Continue down each column.

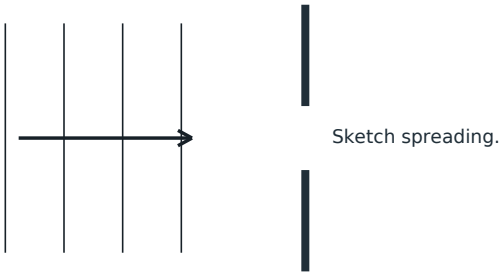
05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise? Explain using wave properties.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed wave speed, frequency becomes 4 times as large. Compare wavelength and typical diffraction through the same opening.
Label wavelength and opening width; sketch the emerging wavefronts.



Marzano check - circle: 1 2 3

07 Exit check

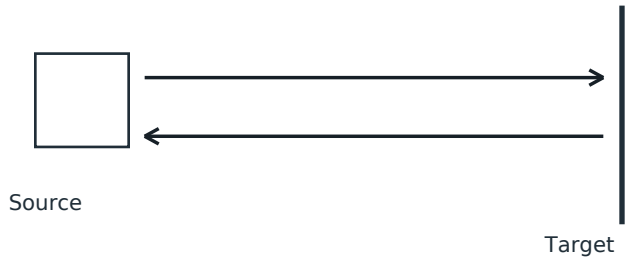
An ideal blackbody temperature doubles while a sound source frequency doubles. These are separate cases. Compare each wavelength change under its appropriate assumptions.

Case	Quantity doubled	Wavelength comparison
Thermal spectrum	Temperature (K)	____
Sound, fixed speed	Frequency (Hz)	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A stationary radar measures echoes at 10.0 and 9.6 microseconds separated by 0.2 s. Find both ranges and average radial velocity; distinguish this from the propagation speed.



Marzano check - circle: 1 2 3